MILITARY SPECIFICATION

FILLING UNIT, CHEMICAL TANK,

MARK 3, MOD - O

1. SCOPE

1.1 This specification covers a portable chemical filling unit.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-E-527	-	Enamel, Alkyd, Lusterless.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Fiberboard.
PPP-T-60	-	Tape; Pressure Sensitive Adhesive, Waterproof - For Packaging and Sealing.
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (For Carton Sealing).

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MIL-C-104	-	*Crates, Wood; Lumber and Plywood Sheathed,* Nailed and Bolted.
MIL-P-116	-	Preservation, Methods of.
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-13811	-	Varnish, Waterproofing, Electrical, Ignition.
MIL-H-15424	-	Hand Tools; Packaging of.

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage.

DRAWINGS

DEPARTMENT OF THE NAVY - BUREAU OF NAVAL WEAPONS

LD2486429	-	Filling Unit, Chemical Tank, Mark 3, Mod - O.
		(All drawings referenced in this specification are listed on Dwg LD2486429.)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

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(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing LD2486429 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the unit shall conform to the specifications and drawings listed on Drawing LD2486429 and subsidiary drawings.

3.2 Assembly. The unit shall be assembled in accordance with Drawing 2486430.

3.3 Unit. The unit, completely assembled less the one ton container, spray tank, filter assembly (Dwg 2486434) and drum of lubricant, shall not leak when subjected to an air pressure of 50 ± 5 pounds per square inch (psi) when tested as specified in 4.4.2.

3.4 Hose assemblies. The hose assemblies (Dwgs 2486480, 2486484) shall withstand one and two-thirds of their respective working pressures without failure when tested as specified in 4.4.1.

3.5 Workmanship. The unit shall be free from contamination (foreign matter, oil or grease) and damage such as dents, cracks or deformation which could affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Special provisions.

4.2.1 Alternative inspection. The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2.3 Preproduction. A preproduction pack shall be examined and tested (see 5.3.1.5) in accordance with 4.4.3.

4.3 Inspection provisions.

4.3.1 For examination and nondestructive tests. Each filling unit shall be examined and tested in accordance with the list of defects.

4.3.2 List of defects.

4.3.2.1 Container stand (Dwg 2486451).

1. Component missing or incorrectly assembled
2. Weld missing or incomplete
3. Saddle radius incorrect
4. Saddle location incorrect
5. Assembly number for matching components missing or does not correspond as specified
6. Holes misaligned

4.3.2.2 Filling unit (Dwg 2486430).

1.	Component missing or incorrectly assembled	
2.	Thread incorrect	
3.	Hose test failure	4.4.1
4.	Unit test failure	4.4.2
5.	Workmanship	3.5

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4.4 Tests.

4.4.1 Hose assemblies. The hose assemblies shall be subjected to the specified pressures + 5 pounds per square inch and examined for leakage during a period of 30 ± 5 seconds. The hose assemblies shall then be subjected to one-half inch of mercury absolute pressure and examined for collapse during a period of 30 ± 5 seconds.

4.4.2 Unit. The unit shall be subjected to the specified pressure and a soap solution applied to all the connections. The unit shall be examined for leakage during a period of 30 + 5 seconds.

4.4.3 Preproduction tests.

4.4.3.1 Impact tests. To determine if a pack is adequately blocked, braced, and anchored, and that provision has been made against shifting of contents within the pack, one of the following tests shall be made: The railroad car method shall be used whenever such facilities are available. The pendulum-impact method shall be used only when facilities for the railroad car method are not available.

4.4.3.2 Railroad car method. The pack shall be securely blocked to prevent movement on the car. Antiskid plates shall not be used for this test. The car shall be humped or shunted to cause impact at approximately 10 miles per hour.

4.4.3.3 Pendulum-impact method. The pack shall be raised with a crane and slings or grabhooks or otherwise suspended from a height sufficient to allow the pack to swing clear of the ground. The pack shall be pulled back and up along the arc of free swing until the center of gravity reaches a point approximately 9 inches above its original position (also point of impact) measured vertically. The pack shall be released and permitted to swing freely so that one end at corner strikes against a solid barrier. The test shall be repeated for the opposite end of the pack.

4.4.3.4 Examination, redesign, and retest. After the preproduction pack test, the pack shall be examined for conformance with 5.3.1.4. Any deficiencies shall be corrected by redesign and the pack retested for conformance with this specification.

5. PREPARATION FOR DELIVERY

5.1 Cleaning and drying. The filling unit and components thereto shall be cleaned and dried in accordance with MIL-P-116, using cleaning process C-1 and the most applicable of drying procedures listed therein.

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5.2 Preservation and packaging. Level shall be as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Matchmarking of disassembled parts. Parts removed by disassembly and parts remaining on the items shall be identically marked by stenciling with either letters or numerals, using lusterless white enamel, conforming to TT-E-527. Parts removed that are too small to accommodate stencil marking, shall have the matchmarking identification shown on tags meeting the requirements of MIL-STD-129.

5.2.1.2 Preservatives and packaging methods. Unless otherwise specified, preservatives, including the application thereof, and packaging methods shall be in accordance with MIL-P-116.

5.2.1.3 Unpainted metal surfaces. Unless otherwise specified, unpainted exposed non-critical and non-functioning surfaces, including threaded areas, shall be coated with type P-2 preservative compound. This preservative compound shall not be applied to stainless steel surfaces.

5.2.1.4 Tape and tape application. Unless otherwise specified, pressure-sensitive tape shall conform to type III, class 1 of PPP-T-60.

5.2.1.5 Unit packaging.

5.2.1.5.1 Reactor. The electric motor shall be removed from the reactor and packaged in accordance with method IIb of MIL-P-116. The inner container shall be a type I, class 1, style RSC, fiberboard box, conforming to PPP-B-636, and the outer container shall be a V3c, style RSC, fiberboard box, conforming to the same specification. In addition to closure requirements specified, the outer container shall have all seams, including the manufacturer's joint, taped, using pressure-sensitive tape, conforming to PPP-T-76. All openings into the reactor shall then be covered with pressure-sensitive tape, overcoating the tape with electrical ignition waterproofing varnish, conforming to MIL-V-13811; or such openings shall be sealed with fitted plastic caps or plugs.

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5.2.1.5.2 Scales. The scales shall be disassembled in accordance with the manufacturer's recommendations. The critical surfaces of the components and parts removed shall be coated with type P-7 preservative oil. Components and parts removed shall be packaged in accordance with method IC-5, except that the removed critical components or parts shall be packaged in accordance with method IA. The above packaged components and parts shall then be packaged together in accordance with method IC-5, using a double walled, V3c, style RSC, fiberboard box as the container. Pressure-sensitive tape conforming to PPP-T-76, shall be used to waterproof the box seams and joints. Wood blocks shall be placed between the frame and platform; and all portions of scale not affected by disassembly, that have parts free to move or vibrate, shall be securely locked or blocked in a fixed position to prevent movement or vibration in transit.

5.2.1.5.3 Pump and motor assembly. This assembly shall be packaged in accordance with method IIa, using an overseas type or class wood box, conforming to either PPP-B-601 or PPP-B-621, as the container.

5.2.1.5.4 Stand, cradle and grab beam. The stand shall be completely disassembled. The hood mechanism on the grab beam shall be oiled with type P-7 preservative. The cradle requires no processing.

5.2.1.5.5 Hose assemblies. Hose fittings shall be cushion-wrapped with grade A greaseproof barrier material, conforming to MIL-B-121, or fitted with plastic caps or plugs. Hose lengths shall then be coiled, with coils secured in at least five places with minimum 1/2-inch wide flat cotton tape. The coiled hose shall then be packaged in accordance with method IC-5. Pressure-sensitive tape, conforming to PPP-T-76, shall be used to waterproof the box seams and joints.

5.2.1.5.6 Plumbing fixtures and associated hardware. Gages shall be packaged in accordance with method IA-15. Valves shall have interior surfaces coated with type P-7 preservative oil and packaged by the same method as is specified for gages. Fittings, such as elbows, tees, unions and pipe lengths shall be packaged together in accordance with method IC-5. Pressure-sensitive tape, conforming to PPP-T-76, shall be used to waterproof the box seams and joints.

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5.2.1.5.7 Canister. M11 canisters shall be packaged as specified on Drawing C5-41-33.

5.2.1.5.8 Repair parts. Repair parts shall be packaged in accordance with MIL-P-116.

5.2.1.5.9 Tools. Tools shall be packaged in accordance with MIL-H-15424.

5.2.1.5.10 One-ton cylinder. No processing is required on this item.

5.2.1.5.11 Technical publications. Technical publications shall be packaged in accordance with method IC-1 or IC-3.

5.2.1.6 Intermediate packaging.

5.2.1.6.1 Canisters. Canisters unit packaged as specified in 5.2.1.5.7 shall be intermediately packaged together in accordance with method IC-5.

5.2.1.6.2 Plumbing fixtures and associated hardware, repair parts, tools and technical publications. These items, unit packaged as specified in 5.2.1.5.6, 5.2.1.5.8, 5.2.1.5.9, and 5.2.1.5.11 shall be intermediately packaged together in accordance with method IC-5, using a double-walled V3c, style RSC, fiberboard box as container.

5.2.2 Level C. The filling unit, including components, repair parts, tools and technical publications shall be packaged to afford adequate protection from the supply source to the first receiving activity for immediate use.

5.3 Packing. Level shall be as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Reactor. The reactor and electric motor processed as specified in 5.2.1.5.1 shall be packed in a type II, class 1 or 2, style a wood crate, conforming to MIL-C-104. The reactor and the packaged electric motor shall each be anchored to the skid base of the crate in accordance with the applicable provisions of the appendix thereto.

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5.3.1.2 Miscellaneous items of filling unit and components, including repair parts, tools, and technical publications. These items unit and intermediately packaged as specified in 5.2.1.5.2, 5.2.1.5.3, 5.2.1.5.4, 5.2.1.5.5, and 5.2.1.6 shall be packed together in a type II, class 1 or 2, style a, wood crate, conforming to MIL-C-104, anchoring these items to the skid base in accordance with the applicable provisions of the appendix thereto.

5.3.1.3 One-ton cylinder. This item with shipping bonnet secured in place shall be shipped loose without any packing.

5.3.1.4 Performance. The filling unit packaged as specified in 5.3.1.1 and 5.3.1.2 shall be capable of withstanding one of the tests described in 4.4.3 without shifting of contents or loosening or the breaking of tie downs and blocking.

5.3.1.5 Preproduction pack. Prior to packing the filling unit in crates in quantity, a preproduction pack shall be prepared and tested to determine conformance with the packing requirements of this specification, except that the pilot pack shall not be required when the vendor has previously had such a pack accepted for identical equipment on a military contract and can furnish satisfactory evidence that the equipment is being packed identically with the approved pack. Approval of the preproduction pack shall not relieve the contractor of his obligation to preserve, package, and pack the equipment in conformance with this specification.

5.3.2 Level C. The filling unit, including components, repair parts, tools and technical publications preserved and packaged in 5.2.2 shall be packed in such manner as to insure carrier acceptance and safe delivery at first domestic destination. Containers shall comply with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.4 Marking. In addition to any special marking required by the contract or order, unit and intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. This item is intended for filling the aero 14 B spray tank with chemical agent. This equipment is used as a portable field unit with a capability of filling the spray tank with either of two chemical agents.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of packaging and packing required.

6.3 Repair parts packaging. When repair parts packaging is required for items authorized for stocking as repair parts, the repair parts packaging data sheets covering the applicable components of the filling unit should be used. (Copies of repair parts packaging data sheets may be obtained from the Director for Engineering and Industrial Services, Attn: Chief, Design Branch, US Army Edgewood Arsenal, Edgewood Arsenal, Maryland, 21010.)

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1040-A026

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8378

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